



First/Second Semester B.E/B.Tech. Degree Examination, June/July 2024
Chemistry for EEE Stream

Max. Marks:100

- Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. M : Marks , L: Bloom's level , C: Course outcomes.
 3. VTU databook is permitted

Module – 1			M	L	C
1	a.	Explain classification of materials as conductors, insulators and semiconductors with the help of band theory.	7	L2	CO1
	b.	Explain the preparation, properties and commercial applications for grapheme oxide.	7	L2	CO1
	c.	Describe the purification of electronic grade silicon from quartz by float zone method.	6	L2	CO1
OR					
2	a.	What are conducting polymers? Explain the mechanism of conduction in polyethylene.	7	L2	CO1
	b.	What is electroless plating? Describe the electroless plating of copper in the manufacture of double-sided PCB.	7	L2	CO1
	c.	A polymer has the following composition 100 molecules of molecular mass 1000 g/mol, 200 molecules of molecular mass 2000g/mol, and 500 molecules of molecular mass 5000g/mol. Calculate the number and weight average molecular weight.	6	L3	CO1
Module – 2					
3	a.	What are Batteries? Explain the classification of batteries with suitable examples.	6	L2	CO2
	b.	Explain the construction and working of sodium-ion battery. Mention its applications.	7	L2	CO2
	c.	Explain the construction and working of vanadium flow battery. Mention its applications.	7	L2	CO2
OR					
4	a.	What are photovoltaic cells? Describe the construction and working of a PV cell. Mention its advantages and disadvantages.	7	L2	CO2
	b.	What are fuel cells? Explain the construction and working of methanol – oxygen fuel cell.	6	L2	CO2
	c.	Explain the construction and working of lithium – polymer battery. Mention its application.	7	L2	CO2
Module – 3					
5	a.	Define corrosion? Explain the electro chemical theory of corrosion taking iron as an example.	7	L2	CO3
	b.	Explain the differentiate metal differential aeration corrosion with an example.	7	L2	CO3
	c.	Calculate the CPR in both MPY and MMPY for a thick steel sheet of area 100 inch ² which experience a weight loss of 485g after one year. (Density of steel = 7.9g/cm ³).	6	L3	CO3

OR

6	a.	What is anodizing? Explain anodizing of aluminium. Mention its application.	7	L2	CO3
	b.	Write a note on : i) Galvanizing ii) Sacrificial anode method.	7	L2	CO3
	c.	What is e-waste? Describe the effects of e-waste on environment and human health.	6	L2	CO3

Module – 4

7	a.	Describe the synthesis of nano-materials by sol-gel method with example.	7	L2	CO4
	b.	Write a note on nanofibers and nanosensors.	7	L2	CO4
	c.	What are QLED? Mention its properties along with their applications.	6	L2	CO4

OR

8	a.	Describe the synthesis of nano-materials by co-precipitation method with an example.	7	L2	CO4
	b.	What are nano-materials? Explain any two size dependent properties of nano-materials.	7	L2	CO4
	c.	What are OLED's? Mention its properties and applications.	6	L2	CO4

Module – 5

9	a.	What are reference electrodes? Explain the construction and working of calomel electrode.	7	L2	CO3
	b.	Explain the working principle and applications of conductometric sensor.	7	L3	CO3
	c.	What are concentration cells? A concentration cell is constructed by immersing two iron electrodes in 0.01m and 0.1m Fe SO ₄ solution represent the cell and calculate EMF of the cell at 298K.	6	L3	CO3

OR

10	a.	What are ion-selective electrodes? Explain the construction and working principle of glass electrode.	7	L2	CO5
	b.	Explain the working principle and applications of colorimetric sensor.	7	L3	CO5
	c.	Explain how the P ^H of the given solution is determined using glass electrode.	6	L2	CO5
